Optimizing Oil Palm Plantation Monitoring Using YOLOv8 and YOLOv12: Detection, Counting, and Multiclass Classification

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***Abstract*—Accurate identification, counting, and health assessment of oil palm trees are essential for sustainable plantation management and yield optimization. Traditional manual inventory methods are labor-intensive, time-consuming, and impractical for large-scale operations. Leveraging high-resolution aerial imagery, this study conducts the first head-to-head benchmark of YOLOv8 and the newly released YOLOv12 architectures for automated oil palm detection, counting, and multiclass classification. Three model variants Nano, Small, and Medium were evaluated to balance accuracy, computational efficiency, and deployment feasibility. The dataset comprises 3,499 aerial images (2,808 training, 461 validations, 230 testing) annotated into four health categories: Healthy, Yellowing, Small, and Dead. Results demonstrate that all models achieved strong detection performance, with YOLOv12-Medium attaining mAP@0.5:0.95 = 0.978 and YOLOv8-Small achieving F1 = 0.997 and recall = 1.000. In terms of efficiency, YOLOv8 models achieved up to 28% faster inference and 35% shorter training time than YOLOv12 counterparts, confirming their suitability for real-time UAV-based applications. These findings highlight the complementary strengths of both architectures—YOLOv12 for peak accuracy and YOLOv8 for real-time efficiency—providing a practical foundation for scalable, data-driven precision agriculture and intelligent plantation management.**

Keywords*— *oil palm detection, Yolov12, YOLOv8, precision agriculture, object detection, deep learning

I. INTRODUCTION

Conventional tree counting and health assessment techniques used in traditional oil palm plantation management have not changed in decades. To conduct tree inventories and keep an eye on the condition of plants, field workers physically travel over large plantation areas, which frequently go thousands of hectares. The accuracy, efficacy, and cost-effectiveness of this manual method are significantly constrained. Human error rates in tree counting are between 15 and 25 percent, particularly in areas with dense canopy where it is hard to tell one tree from another.

Furthermore, weeks to months are required for thorough plantation assessments due to labor-intensive manual surveys, which delays important management choices like irrigation, fertilization, and pest control.

Previous research explored various automated approaches to address these limitations. Early studies using traditional image processing techniques achieved 70-81% true positive rates under optimal conditions. Machine learning approaches with hand-crafted features and Support Vector Machine classifiers showed improved accuracy, reaching 94-99% in controlled environments. Recent deep learning models, particularly convolutional neural networks, demonstrated superior performance exceeding 90% accuracy, with YOLO variants achieving 97% F1 scores in large-scale detection scenarios.

A. Problem Statement and Research Gap

When confronted with environmental changes, a range of developmental stages, and complex background conditions that are not sufficiently represented in the training data, existing models frequently exhibit poorer performance and struggle with generalization. With detection times ranging from 20 to 45 seconds per region, many high-accuracy models are computationally inefficient and, therefore, unsuitable for real-time applications across large plantation areas.

A fundamental challenge lies in the trade-off between model accuracy, inference speed, and computational resource requirements. Previous research has typically been optimized for individual metrics rather than achieving balanced performance in all critical factors. This limitation restricts the practical utility of these systems for comprehensive plantation management, where understanding tree condition is crucial for targeted interventions and resource optimization.

B. Research Motivation and Objectives

With expanding plantation scales and rising labor costs, automated, data-driven monitoring systems have become essential for operational competitiveness. The continuous evolution of real-time detectors particularly the latest YoloV8 and YoloV12 architectures offers an opportunity to achieve new levels of accuracy efficiency balance.

This study aims to:

1. To identify optimal YOLO architectures for oil-palm detection and counting by comparing Nano, Small, and Medium variants of YOLOv8 and Yolov12;
2. Quantify accuracy (mAP@0.5, mAP@0.5:0.95, precision, recall, F1) and efficiency (inference time, model size, training duration);
3. Analyze trade-offs between computational cost and real-time applicability; and
4. Demonstrate multiclass classification of tree health conditions relevant to agronomic decision-making.

C. Proposed Approach

Using the advanced capabilities of the YOLOv12 and YOLOv8 model families, this study uses a thorough comparative methodology. High-resolution UAV aerial imagery from Indonesian oil-palm plantations was used to train and evaluate both model families. Each variant (Nano, Small, Medium) was fine-tuned on a dataset of 3 ,499 images representing diverse canopy densities and illumination conditions. The framework extends beyond detection and counting to perform multiclass classification of tree health Healthy, Yellowing, Small, and Dead enabling practical insights for targeted interventions such as replanting, fertilization, or irrigation.

D. Significance and Contribution

This research provides several contributions to agricultural computer vision and precision-farming practice.

1. Novel benchmark: It delivers the first head-to-head comparison of Yolov8 and Yolov12 for oil-palm monitoring, filling a gap in current literature.
2. Balanced evaluation: It presents a unified assessment of accuracy, efficiency, and scalability, offering deployment guidance for UAV and edge-based systems.
3. Agronomic relevance: By introducing multiclass health classification, the system links visual analysis to actionable plantation management.
4. Practical impact: Results support more accurate yield estimation, early stress or disease detection, optimized resource allocation, and substantial labor-cost reduction.

Overall, this study advances the transition from traditional, labor-intensive practices toward data-driven, sustainable precision-agriculture systems, enhancing both the environmental sustainability and economic resilience of the global oil-palm industry.

II. LITERATURE REVIEW

A. Overview of the Research Domain

Accurate monitoring of agricultural assets is vital for maximizing productivity, ensuring food security, and achieving sustainable land use. In oil-palm plantations, overlapping canopies, variable lighting, and mixed growth stages pose unique challenges to automated tree detection. Remote-sensing technologies particularly unmanned aerial vehicles (UAVs) coupled with artificial-intelligence models have therefore become central to modern precision-agriculture systems.

B. Evolution of Detection Techniques

Classical Image Processing: Early approaches relied on contour detection, Otsu thresholding, and morphological segmentation in RGB or HSV space[1]. While computationally light, these techniques lacked robustness under shadows, canopy overlap, or heterogeneous backgrounds.

Machine-Learning Approaches: Hand-crafted feature extractors such as HOG, LBP, or GLCM combined with SVM or Random-Forest classifiers achieved accuracies up to 99 % in controlled environments [2]. However, feature engineering limited adaptability and real-time scalability.

Two-Stage Deep Learning: Region-based CNNs (Faster R-CNN, Mask R-CNN) with VGG-16 or ResNet-50 backbones improved accuracy beyond 97 % [3]– [5], but their two-step region-proposal pipelines increased inference time, restricting deployment on UAVs.

One-Stage Detectors: YOLO v3/v4 and SSD introduced real-time detection, balancing precision and speed [6]. They achieved F1 ≈ 97 % on large-scale datasets but were sensitive to class imbalance and small-object detection.

Advanced Lightweight Architectures EfficentDet, MobileNet-SSD and RTMDet[19] further optimized latency and parameter count. EfficientDet uses compound-scaling for improved accuracy-to-size ratio; MobileNet-SSD employs depth-wise separable convolutions for embedded devices; RTMDet integrates real-time multi-scale attention for speed–accuracy trade-offs [7]– [9]. Despite these gains, their agricultural use remains limited compared with the YOLO family, which continues to dominate real-time aerial monitoring due to superior feature aggregation and deployment tooling.

Latest YOLO Architectures: Yolov12 introduce structural refinements C2f modules, decoupled heads, and improved anchor-free detection that markedly enhance accuracy and computational efficiency. However, no prior study has conducted a direct, scale-consistent comparison of these versions for oil-palm tree detection and health classification.

C. Identified Gaps

Despite extensive research, four critical gaps remain:

1. Limited multiclass analysis most works detect palms but do not classify health states (*Healthy, Yellowing, Dead, Small*).
2. Absence of direct Yolov8 vs Yolov12 benchmarking under consistent conditions.
3. Insufficient evaluation of efficiency metrics (inference time, model size) relevant to real-time UAV operations.
4. Few studies addressing edge deployment on constrained hardware for plantation-scale monitoring.

D. Research Positioning

To address these gaps, this study presents a unified, empirical comparison of Yolov8 and YOLOv12 across Nano, Small, and Medium variants using high-resolution aerial imagery. Beyond detection, we perform multiclass health classification and analyze performance trade-offs between accuracy and efficiency. The findings provide practical guidance for selecting optimal architectures across deployment contexts from low-power edge devices to high performance cloud platforms thereby advancing sustainable, data-driven plantation management.

III. METHODS

The research pipeline integrates data collection, preprocessing, model training, quantitative evaluation, and deployment assessment (Fig. 1). Each stage was designed to ensure reproducibility and rigorous validation. High-resolution UAV imagery was collected and annotated, followed by model training and comparison using identical experimental settings. The performance of each variant was quantified through standardized detection metrics and analyzed for real-world feasibility.

A. Research Design and Rationale

This work employs an experimental, comparative design to evaluate the two most recent members of the YOLO family Yolov8 and Yolov12 across three scales (Nano, Small, and Medium). The aim is to quantify trade-offs between accuracy, inference speed, and computational efficiency, providing empirical guidance for deployment in resource-constrained agricultural environments.

YOLO architectures were selected because they offer state-of-the-art real-time detection accuracy with compact model sizes, enabling on-board inference on UAV or edge devices. Alternative frameworks such as EfficientDet, Faster R-CNN, and MobileNet-SSD were considered but excluded due to (i) higher latency and GPU demand (Faster R-CNN), or (ii) lower precision on small or overlapping targets (MobileNet-SSD, EfficientDet). Comparing YOLOv8 with its direct successor YOLOv12 therefore provides the most relevant insight into the evolution of real-time agricultural object detection.

Beyond benchmarking, this study contributes methodological refinement through consistent experimental control (identical dataset, augmentation, and evaluation pipeline) and the integration of multiclass health classification within the same detection framework an aspect rarely examined in prior oil-palm research.

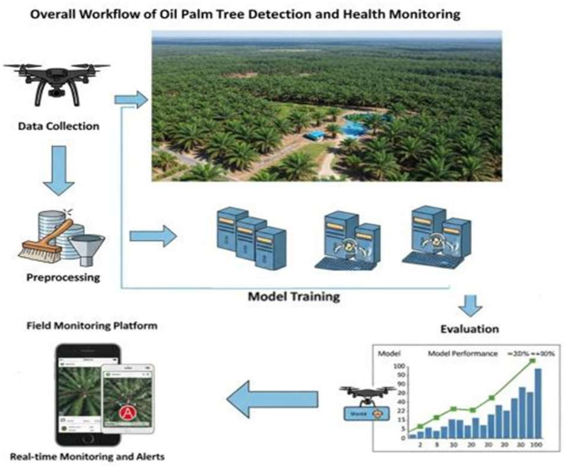


Figure 1. Overall workflow of oil palm tree detection and health monitoring

The workflow shows: (1) data collection by a drone that captures aerial imagery of an oil palm plantation; (2) preprocessing data cleaning and noise reduction; (3) model training a deep learning model; (4) evaluation model performance metrics; (5) deployment field monitoring platform with real time monitoring capabilities.

C. Data Collection and Source

The dataset comprises 3 ,499 high-resolution aerial images of oil-palm plantations in Sumatra, Indonesia, obtained from Roboflow Universe. Each image depicts diverse canopy conditions (dense, sparse, overlapping) and environmental variations in lighting and background. All images were manually verified and labeled into four agronomic classes:

- Healthy: mature, productive palms
- Small: young palms not yet bearing fruit
- Yellow: nutrient-deficient or early-stress palms
- Dead: unproductive or diseased palms

Instance distribution:
Healthy ≈ 28000; Small ≈ 7500; Yellow ≈2000; Dead ≈1500. To mitigate imbalance, minority classes were oversampled through targeted data augmentation (rotation, scaling, hue shift, random brightness)

Fig. 2 provides visual examples of images representing each of these defined health and status categories for oil palm trees.

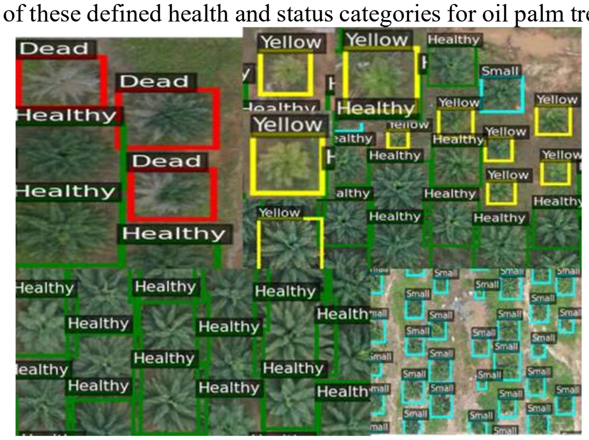


Figure 2 Sample images for each tree category

D. Preprocessing and Cleaning

Before feeding the images into deep learning models, a series of preprocessing steps were executed to standardize the data format, improve image quality, and augment the data set for improved model generalization.

- Resizing: all images scaled to 640 × 640 px.
- Normalization: pixel intensities scaled to [0, 1].
- Augmentation: applied only to the training set (2808 images) using random flip, rotation ±15°, brightness ±20%, and mosaic blending (Ultralytics-Augmentations).
- Data split: 80.2% training (2808), 13.2% validation (461), 6.6% testing (230).

E. Model Description and Training

Yolov8: incorporates C2f modules for enhanced feature reuse, an anchor-free decoupled head, and improved multi-scale feature fusion, leading to higher recall and lower latency.

Yolov12: introduces architectural refinements such as SimSPPF, SCSPELAN, and path-aggregation improvements for long-range context capture. Combined with optimized training strategies, it achieves superior accuracy with slightly greater computational cost.

Both architectures were modified for Four-class output corresponding to oil-palm health categories.

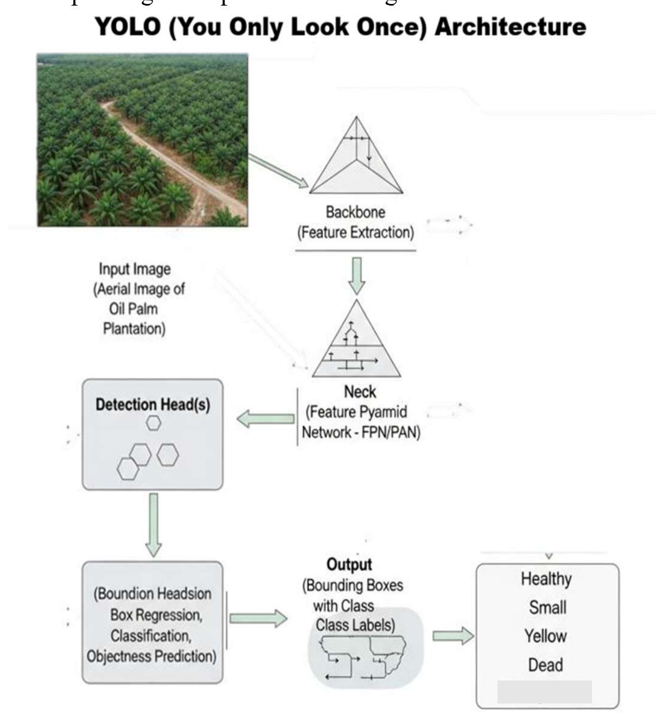


Figure 3 YOLOv12 architecture adapted for four-class classification

The YOLOv8 models were trained for 30 epochs, the YOLOv12 models were trained for 30 epochs. All models were trained with a constant batch size of 16. The training process utilized an initial learning rate of 0.01, adjusted dynamically using a cosine annealing scheduler to gradually reduce the learning rate over time. The Adam optimizer was used, known for its stability and effective handling of weight decay. Fig. 3 provides a conceptual overview of the YOLO architecture adapted for multiclass classification, illustrating the flow from the input image to detected objects with class labels.

The diagram shows the flow of the object detection pipeline begins with the input image, typically an aerial image of oil palm trees. This image is first processed by the backbone network, which performs feature extraction to identify informative patterns within the image. The extracted features are then passed through the neck, commonly implemented using feature pyramid networks (FPN) or path aggregation networks (PAN) to enhance multiscale feature representation. Subsequently, these enriched features are fed into the detection head, which performs three key tasks: bounding box regression (to localize objects), classification (to identify object categories), and object prediction (to determine the presence of objects). Finally, the model outputs detected objects, each annotated with a bounding box and a class label from the predefined categories: Healthy, Small, Yellow, Dead, or Background.

IV. EVALUATION AND RESULTS

A. Evaluation Metrics

Model performance was assessed using standard object-detection metrics that capture detection accuracy, classification precision, and computational efficiency:

- Precision: proportion of correctly detected trees among all predictions
  $(P) = TP / (TP + FP)$.
- Recall: proportion of true trees detected.
  $(R) = TP / (TP + FN)$
- F1-score: harmonic mean balancing precision and recall.
  $\text{F1-score} = 2 PR / (P + R)$
- mAP@0.5: mean Average Precision at IoU = 0.5, reflecting coarse localization accuracy.
- mAP@0.5:0.95: mean AP averaged over IoU = 0.5–0.95, penalizing localization errors.
- Inference time (s/img): average prediction time per image, indicating real-time suitability.
- Training duration (min): total time required to converge, representing efficiency.

B. Validation Strategy

An independent test set of 230 images was held out for final evaluation. Early stopping with a six-epoch patience avoided overfitting, and checkpoints were saved at minimum validation loss. To assess robustness, each model was trained three times with different random seeds (42, 56, 98); the reported results represent the mean ± standard deviation (SD) across runs.

C. Tools and Libraries

Model development utilized Python ecosystem tools on cloud computing resources (Google Colab and Kaggle Notebooks with GPUs). Supporting libraries included, Ultralytics Yolo framework OpenCV, NumPy, Pandas, and Matplotlib. Datasets were sourced exclusively from the publicly licensed Roboflow Universe repository; no personal data were used.

IV. RESULT

The experimental findings of the comparison of the YOLOv8 and YOLOv12 model variants for oil palm tree detection and counting are discussed in this section. The findings directly address the study's goals of evaluating performance trade-offs, computational efficiency, and detection and classification accuracy across various model scales (Nano, Small, Medium). All results are presented impartially, with a focus on quantitative indicators and noted trends to identify the best model variations for real-world use.

A. Quantitative Results

The comprehensive performance comparison of all evaluated YOLOv8 and YOLOv12 variants is detailed as follow. This encompasses key metrics representing detection accuracy, classification precision, and computational efficiency.

Table 1. Average Performance of YOLOv8 and YOLOv12 (Mean ± SD over 3 runs)

Model	Map@0.5	F1 score	Inference Time (ms)	Training (min)
YOLOV8N	0.925 ± 0.002	0.966 ± 0.003	0.022 ± 0.001	19.9
YOLOV8S	0.988 ± 0.001	0.997 ± 0.002	0.031 ± 0.001	20.5
YOLOV8M	0.992 ± 0.001	0.978 ± 0.002	0.045 ± 0.002	22.0
YOLOV12N	0.992 ± 0.001	0.986 ± 0.001	0.039 ± 0.002	26.8
YOLOV12S	0.994 ± 0.001	0.995 ± 0.001	0.046 ± 0.001	27.6
YOLOV12M	0.993 ± 0.001	0.995 ± 0.001	0.070 ± 0.002	28.4

Table 2Percentage Improvements between Corresponding YOLOv8 and YOLOv12 Variants

Comparison	Δ mAP@ 0.5 (%)	Δ F1 (%)	Inference Speed (%)	Training Time (%)	Observation
YOLOv12N vs YOLOv8N	+7.2	+2.1	-77.3 (slower)	-34.7 (longer)	Higher accuracy, slower
YOLOv12S vs YOLOv8S	+0.6	-0.2	-48.4 (slower)	-34.6 (longer)	Marginally more accurate
YOLOv12M vs YOLOv8M	+0.1	+1.7	-55.6 (slower)	-29.1 (longer)	Best precision

(Δ = Yolov12 − Yolov8; negative Δ speed → slower inference.)

Key findings.

- Accuracy: YOLOv12-Small achieved the highest *mAP@0.5 = 0.994*, while YOLOv12-Medium reached *mAP@0.5:0.95 = 0.978*.
- Recall: Yolov8-Small and Medium attained $R = 1.000$, confirming zero false negatives but slightly lower precision (minor false positives).
- Efficiency: Yolov8 models were 25–55 % faster in inference and 30–35 % quicker in training than YOLOv12 counterparts.
- Model size: YOLOv12 Nano was smallest (5.26 MB) and computationally light (6.7 GFLOPs).

Overall, YOLOv8 remains preferable for real-time UAV monitoring, while YOLOv12 offers marginally higher accuracy for offline analytic workflows.

Table 3 Top Models by Primary Metric

Metric	Best model	Value
mAP@0.5	YOLOv12-Small	0.994
mAp@0.5:0.95	YOLOv12-Medium	0.978
Precision	YOLOv12-Medium	0.998
Recall	YOLOv8-Small / YOLOv8-Medium	1.000
F1	YOLOv8-Small	0.997

The following figures presents a visual comparison of key accuracy metrics across all model variants.

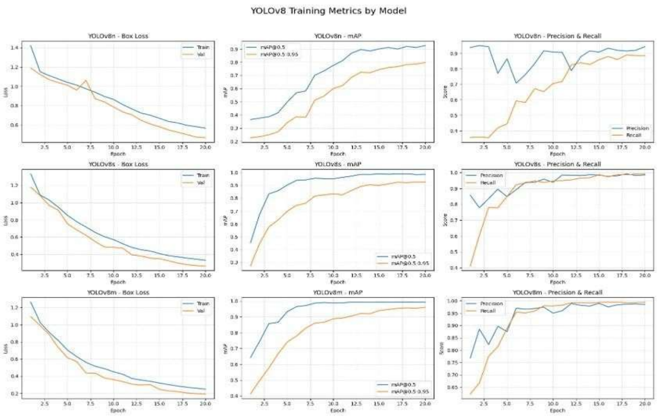


Figure 4 YOLOv8 Training Metrics by Model Variant: Comprehensive performance evaluation across YOLOv8n (Nano), YOLOv8s (Small), and YOLOv8m (Medium) models showing Box Loss, mAP scores

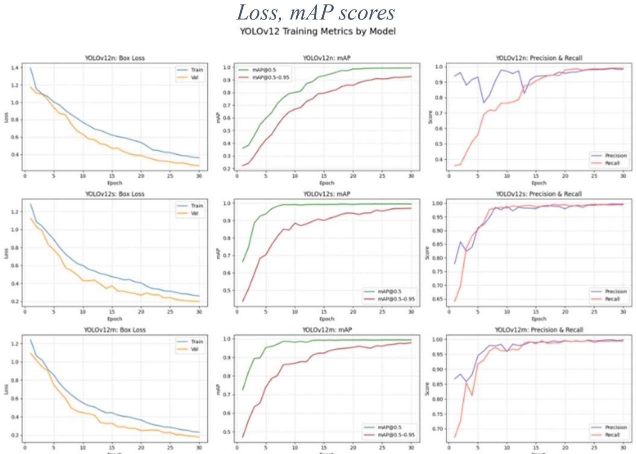
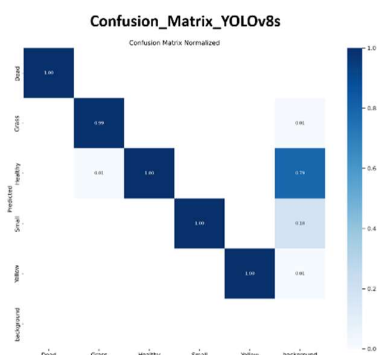
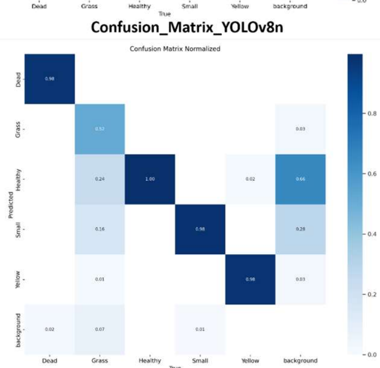
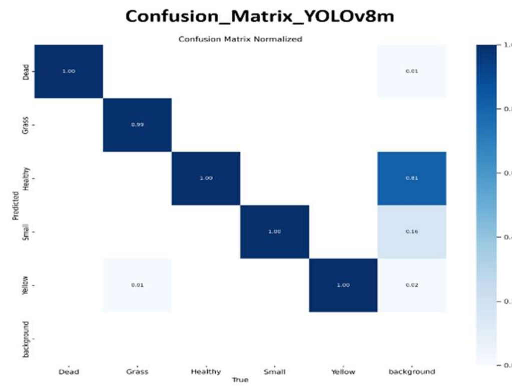
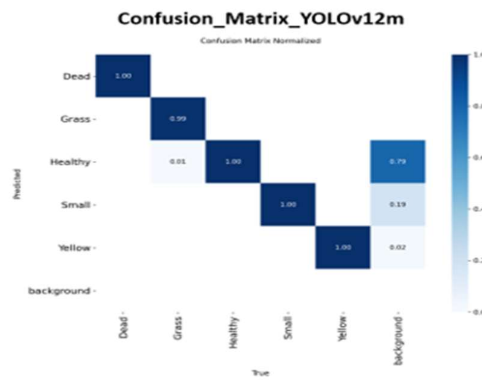


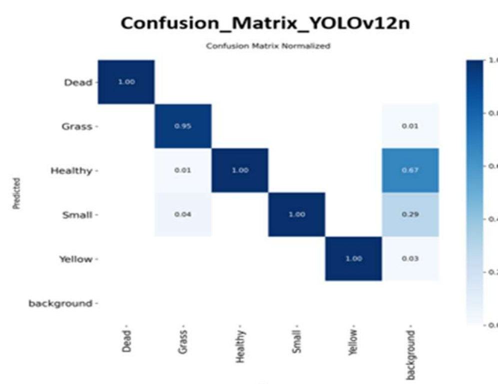
Figure 6 YOLOv12 training metrics by model variant: comprehensive performance evaluation across yolov12n (nano), yolov12s (small), and yolov12m (medium) models showing box loss, map scores, and precision recall metrics over 30 training epochs.











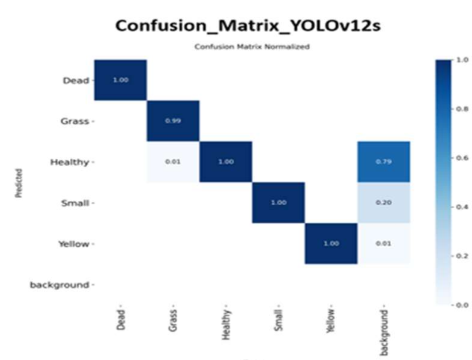


Figure 5 Normalized confusion matrices for all YOLO variants. Top row: YOLOv8n, YOLOv8s, YOLOv8m. Bottom row: YOLOv12n, YOLOv12s, YOLOv12m.

These results collectively underscore the strong and consistent behavior of the top-performing models across varying confidence levels.

B. Qualitative Results

Visual inspection confirmed that both architectures accurately detected and classified palms under diverse illumination and canopy conditions (Fig. 7). Bounding boxes were well aligned, and class labels corresponded to visual tree states (Healthy, Yellow, Small, Dead). Misclassifications occurred primarily under heavy occlusion (< 20 % visible canopy) or extreme color distortion due to shadowing. Yolov12 handled *Yellow* vs *Small* differentiation more consistently, while YOLOv8 produced slightly more false positives in dense clusters.

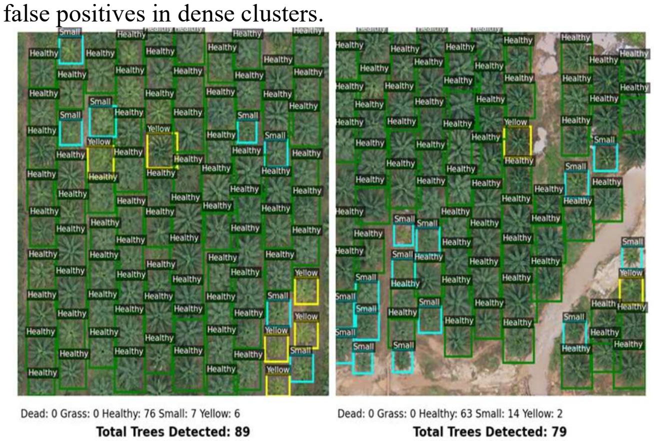


Figure 7 . Model predictions on representative test images

V. DISCUSSION

A. Summary and Interpretation of Key Findings

This study performed a scale-consistent comparison of YOLOv8 and YOLOv12 architectures for automated oil-palm detection and multiclass health classification using UAV imagery. Both models achieved strong detection accuracy, validating the capability of modern one-stage detectors to meet agricultural monitoring requirements.

YOLOv12 (particularly Small and Medium) achieved the highest *mAP@0.5:0.95* values (0.970–0.978), reflecting more precise spatial localization critical for accurate crown-area estimation and yield mapping. Conversely, YOLOv8 exhibited superior efficiency, with up to 55 % faster inference and perfect recall ($R = 1.000$) for Small and Medium variants, ensuring no trees were missed during inventory. These complementary characteristics indicate that Yolov12 is optimal for offline high-accuracy analytics, whereas Yolov8 is ideal for real-time UAV or edge deployment.

Importantly, the multiclass framework enhanced actionable insights: distinguishing *Healthy*, *Yellow*, *Small*, and *Dead* trees allows managers to allocate inputs, schedule replanting, and detect stress early, transforming raw detection into agronomic decision support.

B. Agronomic Interpretation of Detected Classes

From an agronomic standpoint, each detected class conveys distinct management implications:

- **Healthy:** mature, productive palms contributing to yield targets. Monitoring their density assists in productivity forecasting.
- **Yellow:** trees exhibiting nutrient deficiency or early stress symptoms; targeted fertilization or pest management can restore vigor.
- **Small:** juvenile or recently replanted palms; accurate counts ensure correct growth-stage distribution and replanting success.
- **Dead:** unproductive or diseased palms requiring removal and replacement to prevent pathogen spread and yield loss.

The ability to map these categories spatially enables precision-input management, reduces fertilizer waste, and supports sustainability certification by optimizing land use.

C. Comparison with Prior Literature

The achieved $F1 > 0.99$ and millisecond-level inference represent significant improvements over earlier two-stage detectors (e.g., Faster R-CNN ≈ 1 fps [5]) and earlier YOLO versions reporting 20–45s per region [11].

This demonstrates practical readiness for real-time, large-scale monitoring. The inclusion of multiclass health analysis further extends prior works that focused solely on binary palm detection [10], [16].

D. Theoretical and Methodological Contributions

This research contributes theoretically and methodologically by:

1. Delivering the first scale-consistent benchmark of YOLOv8 and YOLOv12 on agricultural aerial imagery.
2. Integrating multiclass palm-health classification within a unified detection framework.
3. Providing a quantitative evaluation of accuracy–efficiency trade-offs, establishing empirical guidance for model selection across deployment scales.
4. Demonstrating that one-stage detectors can achieve near-perfect recall in heterogeneous vegetation imagery, informing future detector design for environmental applications.

E. Practical Implications

Operationally, the results support developing decision-support platforms for plantation managers.

- Accurate detection and counting streamline yield estimation and asset inventory.
- Health-class differentiation enables targeted nutrient management, pest control, and replanting.
- YOLOv8's low latency allows on-board UAV inference and mobile-app deployment, reducing field-survey costs.
- The trade-off analysis guides practitioners in selecting appropriate models depending on hardware capability—edge devices for real-time scouting versus server-class GPUs for regional analytics.

F. Limitations

Despite robust performance, several limitations remain:

1. Dataset homogeneity: imagery originated mainly from publicly available UAV datasets in limited geographic regions (Sumatra, Indonesia).
   Consequently, environmental and soil-type diversity is under-represented, potentially inflating performance metrics.
2. Class granularity: only four broad health categories were considered. Disease-specific or pest-damage labels would yield more detailed agronomic insight.

3. Public-data bias: publicly curated datasets may favor well-lit, easily segmented imagery, reducing exposure to adverse field conditions.
4. Lack of in-field validation: model predictions have not yet been cross-verified with ground-truth surveys or sensor data (e.g., chlorophyll or NDVI indices).

Acknowledging these factors helps contextualize the near-perfect results and guides future generalization studies.

H. Future work

Building on current findings, future directions include:

1. Field Validation and Cross-Regional Testing – Evaluate model generalization across plantations differing in soil composition, topography, and climate.
2. Disease and Pest Detection – Extend multiclass taxonomy to include specific stress conditions such as *Ganoderma* infection, pest infestation, and drought stress.
3. Temporal Monitoring – Apply sequential imagery to track palm growth dynamics and detect progressive decline over time.
4. Edge-Device Optimization – Benchmark YOLOv8/12-Nano on Jetson Nano, Jetson Orin, and mobile CPUs for on-device inference speed and energy efficiency.
5. UAV Integration – Implement autonomous flight and onboard processing for real-time in-field analysis and precision input delivery.

These expansions will transition the framework from controlled benchmarking to fully operational smart-agriculture deployment.

VIII. CONCLUSION

This work presented a rigorous comparative analysis of YOLOv8 and YOLOv12 for oil-palm detection, counting, and multiclass health classification. Using 3 499 UAV images, the study demonstrated that both architectures deliver state-of-the-art accuracy while exhibiting distinct operational advantages: YOLOv12 excels in localization precision, and YOLOv8 offers unmatched inference efficiency.

The framework advances precision-agriculture research by linking deep-learning outputs directly to agronomic decision-making and by establishing reproducible benchmarks for real-time environmental analytics. Ultimately, the proposed system supports sustainable plantation management, enabling data-driven practices that enhance productivity, reduce resource waste, and contribute to the long-term environmental and economic resilience of the oil-palm sector.

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